

Reader Report: Week 12

Thompson Sampling, Latent-Space BO, and Causal Inference from Text

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Summary

BO completion. Thompson sampling draws $\tilde{f} \sim p(f \mid \mathcal{D}_t)$ and maximizes it, providing the first known frequentist sublinear regret bound for nonparametric GP bandits (Chowdhury and Gopalan, 2017), with no free β_t schedule and natural batch parallelism. Latent-space BO places a learned low-dimensional embedding before the GP, making Thompson sampling tractable where UCB optimization degrades.

Causal inference from text. Egami et al. (2022) establish the core framework: text appears as *treatment* or *outcome*, and in both roles using the same documents to discover the representation g and estimate effects creates an identification problem (Analyst Induced SUTVA Violation) and an estimation problem (overfitting), resolved by a train/test split. Veitch et al. (2020) formalize when a learned embedding is causally sufficient: $\hat{Z} = \lambda(\mathbf{W})$ suffices if and only if it captures all of $\mathbf{W}$ predictive of both T and Y (Theorem 3.1), achievable asymptotically by jointly fine-tuning on T and Y (cf. Feder et al., 2022; Sridhar and Blei, 2022). Imai and Nakamura (2024) bypass observational representation learning entirely: an LLM generates treatment texts via prompting, and its *internal representations* are extracted for statistical adjustment, achieving smaller bias and RMSE than observational baselines (cf. Feldman et al., 2026).

Critique

The LLM turn substitutes one untestable assumption for another: “the text $\mathbf{W}$ suffices for causal adjustment” becomes “the LLM representation is *causally sufficient for this domain*,” potentially worse if pretraining under-represents institutional context.

Causal sufficiency, however, has a derivable testable implication. Veitch et al.’s Theorem 3.1 states $\hat{Z} = \lambda(\mathbf{W})$ is causally sufficient iff it captures all of $\mathbf{W}$ relevant to predicting both T and Y . Under a structural causal model, this implies $P(Y \mid T, \hat{Z}, E=e) = P(Y \mid T, \hat{Z})$ for any environment variable E defined by causally irrelevant variation in $\mathbf{W}$ (style, register, petition length): if $\hat{Z}$ fully captures the true confounders $Z = f(\mathbf{W})$, then E adds no additional information about Z beyond $\hat{Z}$. Contrapositive: environment-varying effect estimates — detectable via IRM (Arjovsky et al., 2019) or invariant prediction (Peters et al., 2016) — constitute direct evidence that $\hat{Z}$ has not fully captured Z and is not causally sufficient. This is a consequence of Theorem 3.1 but is not stated in any assigned paper.

The BO connection is more than analogy. Feldman et al. (2026)’s SAE feature space defines a continuous action space $\mathcal{A} \subseteq \mathbb{R}^k$, and the causal effect $\tau(a)$ is an unknown function over $\mathcal{A}$ estimated by the R-learner with standard errors at each action — exactly the noisy observations required for a GP surrogate. Thompson sampling from the resulting GP posterior is then an adaptive design policy for this problem, requiring no additional machinery beyond what each literature has already developed (Aglietti et al., 2021).

Questions

1. Egami et al.’s train/test split halves sample size. Is there a Bayesian analog — a posterior over g propagated into the causal estimand — that avoids this cost? The GP empirical Bayes analogy (Week 11) seems the natural starting point.
2. The invariance test is a *necessary* condition derivable from Theorem 3.1. Does minimizing IRM loss across stylistic environments yield causal sufficiency in the Theorem 3.1 sense, or does the IRM objective (finite discrete environments) strictly weaken the measurability condition?

Project Update: Text as Confounder Proxy and Treatment in the Zoning Data

Abstract

The Austin zoning dataset contains public testimony — oral and written comment — associated with each of the 7,074 cases. This week I formalize how text enters the causal model in two distinct roles and map each role to the appropriate method from the readings.

Two Roles for Text in the Zoning Data

Role 1: Text as confounder proxy. A protest petition signals community opposition, but its *content* reveals the community’s mobilization capacity, political framing, and institutional sophistication — latent confounders for both the petition-filing decision and the council’s vote. Standard covariate adjustment cannot condition on raw petition text. The Veitch et al. (2020) framework applies: learn a causally sufficient embedding $\hat{Z} = g(\text{petition})$ by jointly fine-tuning a pretrained language model to predict (a) whether the case was appealed post-decision and (b) final vote outcome, then include $\hat{Z}$ as a covariate in the GP surrogate for the objective function f .

Role 2: Text as treatment. Whether protest rhetoric appeals to environmental, fiscal, or racial-equity concerns may have a direct causal effect on vote outcome, independent of whether a formal protest petition was filed. The Egami et al. (2022) train/test split applies: (a) use the pre-2022 training set ($n = 4,312$) to learn a topic model or cluster the petition text into rhetorical frames; (b) use the post-2022 test set ($n = 2,762$) to estimate the causal effect of frame on vote, conditioning on $\hat{Z}$ from Role 1. This separation prevents the Analyst Induced SUTVA Violation (AISV) and overfitting.

LLM Augmentation for the Post-2022 Sample

The change-point identified in previous weeks (2022 municipal election) compounds the text problem: the distribution of petition rhetoric and the council’s sensitivity to it may have shifted simultaneously. The post-2022 inference sample is already small ($n = 2,762$) and cannot support reliable topic discovery. The Imai and Nakamura (2024) GPI approach is attractive here: rather than discovering the rhetorical frame from observational text, use an LLM to *generate* treatment petitions via prompting (with the framing condition specified), then extract the LLM’s internal representations of those texts for statistical adjustment of confounding features.

The required assumption — that the LLM’s internal representation captures Austin zoning-specific institutional semantics — is testable via the invariance diagnostic proposed in my critique above: estimate the causal effect of rhetorical frame separately for cases decided by different council committees, and test whether estimates are invariant across committees. Failure of invariance would indicate that the LLM representation conflates framing with committee-specific procedural norms, motivating IRM-style fine-tuning before the GPI estimation step.

Next Steps

(1) Embed petition text using a domain-adapted model (BERT fine-tuned on municipal documents) and perform the invariance test across council committees. (2) Implement the Egami et al. split-sample design for the frame-as-treatment analysis. (3) If the invariance test fails, implement IRM fine-tuning of the embedding as a preprocessing step. (4) Integrate $\hat{Z}$ as a covariate in the GP surrogate from previous weeks and evaluate the change in counterfactual comparison results.

References

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